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► **Algorithm:** given N states at iteration $t - 1$





2. **Communication:** Propose swaps for adjacent components
 - ▶ **Accept** / **Reject** according to Metropolis-Hastings criterion















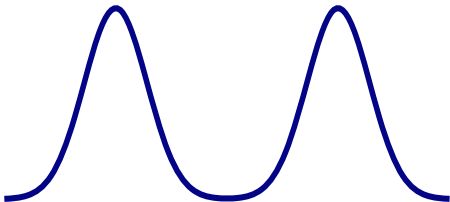




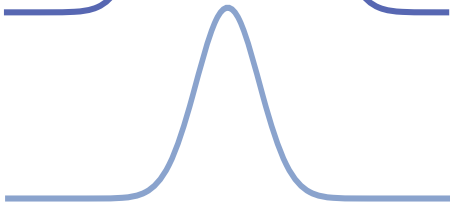
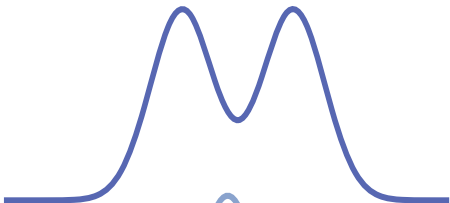








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Target

$$X_t^N$$

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$$X_t^1$$

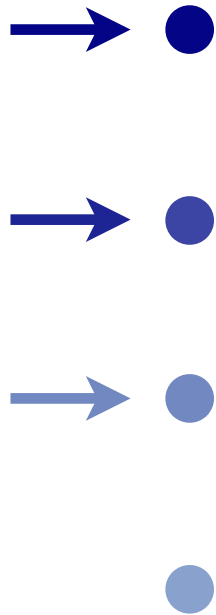
Reference

$$X_t^0$$



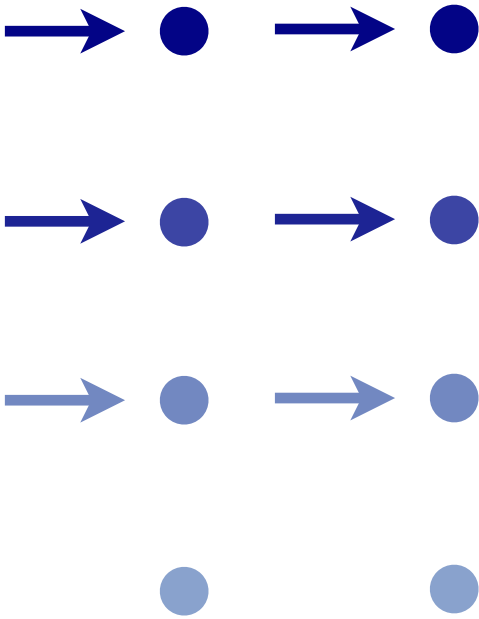
stabilised MCMC target annealing distributions in parallel





1. **Local exploration:** Update n -th component with MCMC move for π_{β_n}



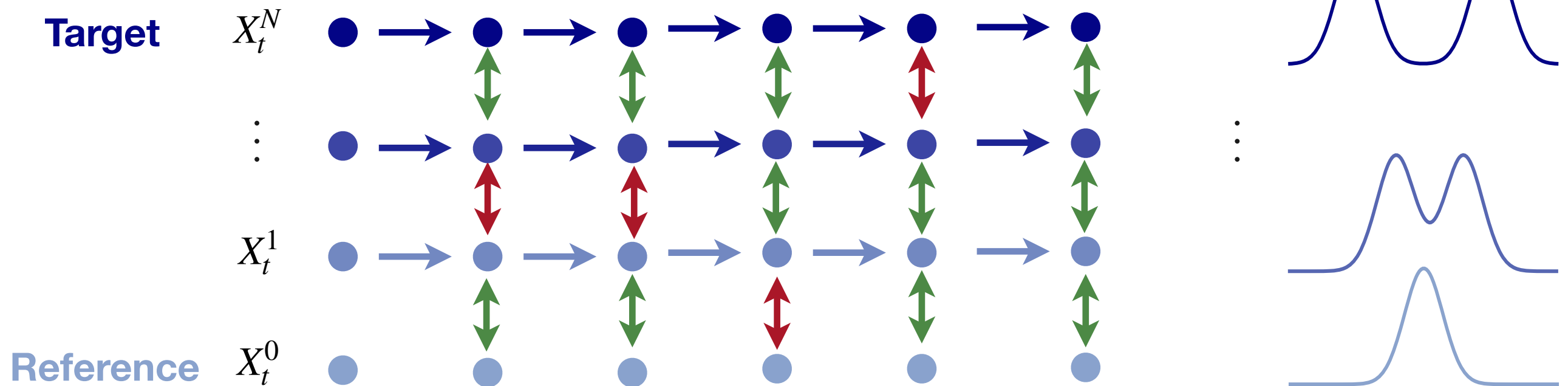




PARALLEL TEMPERING

PARALLEL TEMPERING

- Uses stabilised MCMC to target annealing distributions in **parallel**



- **Algorithm:** given N states at iteration $t - 1$

1. **Local exploration:** Update n -th component with MCMC move for π_{β_n}
2. **Communication:** Propose swaps for adjacent components
 - **Accept / Reject** according to Metropolis-Hastings criterion

PARALLEL TEMPERING